

# Dipesh Balani

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## WORK EXPERIENCES

- **Avis Budget Group** June 2025 - Present
  - *Data Scientist – Pricing*
    - Deployed **ML pricing and demand-forecasting models** (regression, tree-based) estimating price elasticity and booking probability, improving forecast accuracy by **~12%** and powering dynamic rate optimization.
    - Engineered feature-rich datasets from **millions of transactional and fleet records** using **SQL/Python**, incorporating seasonality and demand variation to boost model robustness.
    - Built value-estimation and utilization models driving data-driven capital allocation, and ran scenario/sensitivity analysis to quantify revenue impact and guide pricing strategy.
- **Gainwell Technologies** July 2024 - October 2024
  - *Financial Engineer Intern*
    - Redesigned financial consolidation workflows by integrating **SAP S/4HANA** with **AWS Redshift**, cutting manual reporting hours by **60%** and accelerating month-end close by **27%**.
    - Led variance analysis across **8+ cost centers**, uncovering **\$1.17M in potential savings** and improving forecasting accuracy by **14%**.
    - Boosted cash flow forecasting accuracy by **17.8%** using **Excel (Solver, Power Query) & SAP Analytics Cloud**; delivered risk dashboards to senior stakeholders.
- **University of Maryland** January 2024 - May 2025
  - *Research Assistant, Data Scientist*
    - Built and validated **ML models** (XGBoost, Random Forest, deep learning) on a 500K+ record dataset for a faculty-led research initiative, improving predictive accuracy by **~16%** and shaping methodology adopted in an ongoing peer-reviewed study.
    - Designed **A/B tests** and causal-inference analyses to evaluate model-driven interventions, and built automated **Python/SQL** data pipelines that cut manual processing time by **70%**, accelerating iteration across 3 concurrent research projects.

## EDUCATION

- **University of Maryland** College Park
  - *Master of Science, Information Systems (Applied Quantitative Finance)* August 2023 - December 2024
- **Vellore Institute of Technology** Vellore, India
  - *Bachelor of Science, Computer Science* July 2019 - June 2023

## SELECTED PROJECTS

- **Employee Attrition Prediction & Retention Analytics – IBM HR Analytics:**
  - Built classification pipeline (1,470 records, SMOTE-balanced); LogReg/RF/**XGBoost** reached **0.89 AUC-ROC**.
  - **SHAP**-driven attrition drivers informed interventions projected to save **\$410K/yr**, shipped via a **Streamlit** dashboard.
  - Validated robustness via stratified 5-fold CV and designed an **A/B test** to measure real-world intervention impact before full rollout.
- **Match Outcome Prediction & Stats Chatbot – European Soccer Data:**
  - Built **ETL pipeline** (~25K matches, 40+ features); Poisson/XGBoost/**LSTM** reached **58% accuracy** (vs. 46%).
  - Extended to a **RAG chatbot** (LangChain, vector search) for match/player Q&A at **91% accuracy**.
  - Automated weekly retraining and drift monitoring via an **Airflow** DAG, keeping predictions calibrated across a full season.
- **Chest X-Ray Pattern Analysis – Deep Learning for Medical Imaging:**
  - Built **CNN pipeline** (**ResNet-50/DenseNet-121**, 15K+ X-rays); focal loss lifted sensitivity **61%→89%**.
  - Reached **0.94 AUC-ROC** vs. CheXNet/COVID-Net; **Grad-CAM**-validated, Dockerized FastAPI inference at <200ms.
  - Benchmarked 4 additional architectures (EfficientNet, VGG16, Vision Transformer) and ensembled top performers, improving generalization across held-out hospital sites.

## SKILLS

- **Programming & Data:** Python, R, SQL, Java, NumPy, Pandas, SciPy, PySpark, Git, Linux/Bash
- **Machine Learning & Deep Learning:** Scikit-learn, XGBoost, LightGBM, Random Forest, PyTorch, TensorFlow, Keras, CNN, LSTM/RNN, Transformers, Hugging Face, Transfer Learning, NLP, Computer Vision, SHAP
- **GenAI & MLOps:** LangChain, RAG, Vector Databases (FAISS, Pinecone), Prompt Engineering, Docker, Kubernetes, FastAPI, Airflow, MLflow, CI/CD, AWS (SageMaker, Redshift, Athena), GCP
- **Statistics, Experimentation & Visualization:** A/B Testing, Hypothesis Testing, Causal Inference, Time Series Analysis, Monte Carlo Simulation, Bayesian Methods, Tableau, Power BI, Palantir Foundry